

Assignment 5 Verification and Comparison

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Assessment of the Original Paper

The smart-farming paper already contains sufficient scientific material for this conversion exercise. Assignment 5 explicitly requires preservation of the original science and says to transfer sections wherever applicable. Its example skeleton is not a reason to remove existing headings or fabricate experimental results for a narrative review. No original section, paragraph, equation, table, figure, or reference has been deliberately removed.

Guideline component	Location in the original paper	Assessment
Introduction	Section 1	Present and retained
Related work	Sections 1, 6, and 7	Distributed literature discussion retained
Methodology	Section 2	Narrative-review approach and evidence boundaries retained
Dataset and experimental setup	Sections 2 and 8.3	No new dataset or experiment claimed; proposed protocol retained
Results and discussion	Sections 7 and 8	Literature synthesis and illustrative calculation, not new empirical results
Limitations	Sections 2, 6.2, and 9	Evidence, model, and deployment limits retained
Conclusion	Section 10	Present and retained
Additional technical material	Sections 3-6, 8-9 and Research Transparency	All retained, including governing equations and calibration

Preservation and Additions

The original PDF is included unchanged as Original_Paper.pdf. The conversion retains the exact title, abstract, keywords, 10 numbered main sections, all original subsections, Research Transparency, 12 display equations, two complete tables, one original architecture figure, and 20 manuscript references. R21-R25 from the wider repository are not added to the paper's bibliography.

Document-preparation additions comprise an author/affiliation block from existing AS5 records, dynamic citations and object references, and one navigational section-reference sentence. After the first platform test, the student approved Prism's one-paragraph clarity revision in Section 5.1. All other prose, equations, tables, figures, and references remain unchanged from the prepared conversion. The original PDF is unchanged. BibTeX may change punctuation and citation order while retaining reference identity. All numerical assumptions and the illustrative 200 m3 result are retained; no new measured result or scientific conclusion is introduced.

Consistency Checks

Item	Source representation	LaTeX representation and check
Mathematics	Rendered equation images	12 editable expressions; original operators and signs retained
Inline symbols	Greek letters and sub/superscripts in Word	Corresponding mathematical notation in LaTeX
Units	mm, m3, kg/ha, kg/m3	Same quantities; superscript 3 typeset mathematically
Object numbering	Typed figure, table, and equation numbers	Labels and reference commands
Citations	Numeric text including ranges	R01-R20 citation keys and generated bibliography
Evidence level	Review, numerical study, deployment, and preprint distinguished	Distinctions retained
Water accounting	Application distinguished from consumptive use	Original distinction retained

Paragraph Comparison

Version A Original Paragraph

Calibration estimates model parameters, whereas data assimilation updates the estimated system state as observations arrive. Sensor calibration is a separate measurement problem. These three activities should be documented independently: an apparently successful hydraulic calibration can otherwise compensate for a biased sensor, incorrect irrigation input, or unmodelled water-table influence.

Version B Actual Prism Revision

Model calibration estimates model parameters, data assimilation updates the estimated system state as observations become available, and sensor calibration addresses measurement accuracy. These three activities should be documented separately; otherwise, an apparently successful hydraulic calibration may compensate for sensor bias, incorrect irrigation inputs, or unmodelled water-table influence.

This is the actual response to Prism Task 6, received on 26 September 2026. It supersedes the earlier local candidate.

Version C Student Approved

Version C equals Version B above, without further wording changes. On 27 September 2026 the student explicitly selected "Use Prism's revised paragraph" when asked to choose between the original and Prism revision. This records the student's approval, not an invented claim of independent external scientific validation. Prism Task 7 applied exactly that paragraph to main.tex. The export comparison confirms that no other source content changed.

Analysis

Prism combines the three activity definitions into parallel clauses and replaces "is a separate measurement problem" with "addresses measurement accuracy". The separate-documentation requirement remains, as do sensor bias, incorrect irrigation inputs, and unmodelled water-table influence. "May compensate" retains a possibility rather than claiming a measured effect, though it is a wording change from "can otherwise compensate". The revision is clearer and more compact, but it does not validate calibration or eliminate confounding. The student approved this stylistic tradeoff. The unchanged original remains available as Version A.

Platform Portability Record

Component	Local project	Prism	Overleaf
Compilation	Local preparation passed	Genuine platform PDF, 13 pages	First unchanged export compiled, 13 pages
Sections and abstract	Preserved in main.tex	Retained	Retained
Equations and tables	12 equations, 55 table cells	Native and retained	Native and retained
Figure and asset path	Relative Figures path	Original asset retained	Original asset retained
Citations and references	BibTeX R01-R20	20 resolved entries	20 resolved entries
Cross-references	Four object categories	All source targets present	No undefined-reference warnings

The first genuine Prism export was uploaded unchanged into a new Overleaf project on 26 September 2026. Its SHA-256 is 299a434fb9ce561a4e7cbc8a6fc00908344b00709bdeeb7ded943ff1ad91c074. All three files matched the local conversion byte-for-byte. No first-attempt source or compiler-setting correction was needed. Overleaf showed zero errors, zero warnings, and one typesetting information entry; its retained log identifies an underfull bibliography line. Both first-run PDFs contain identical page text after removing only whitespace and Unicode mathematical variation selectors. Full logs and initial outputs are retained under Evidence.

After student approval, Prism changed only the Section 5.1 paragraph. The final Prism export and downloaded final platform PDFs are separate from the preserved first-run evidence. The original old-topic AS5 platform files remain untouched. Local_Compiled_Paper.pdf is the earlier local preparation build, not a platform output and not the later approved-paragraph version.

Final Student Checks

Final verification on 27 September confirms 13 pages in each final platform PDF, with every page's extracted text identical after whitespace and Unicode variation-selector normalization. The final export SHA-256 is a13a82d6ebb793a634a99e2976c6d151add347d8ba23dd7448ae321d047e53f7. All three exported source files match the submission sources byte-for-byte. The final Overleaf run again shows zero errors, zero warnings, and one minor bibliography-spacing information entry. The comparison is in Evidence/platform-validation.json; final compiler outputs are in Evidence/overleaf-final-run.

The platform session, actual prompts, original first-run test, and student paragraph choice are documented. Before submission, the student should confirm the author block and scientific content, review the reflection as a truthful account of this agent-assisted session, and follow any course rules governing AI assistance. Approval of one paragraph is not treated as approval of the entire manuscript or reflection.